

Report
Of
ONLINE COURSE
&
LEARNING MANAGEMENT SYSTEM

Submitted
To
School of Computer Science & Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B. Tech CSE



By
Roll Number: 25SCS1003000700
Name of Student: **PRATYAKSH BHARDWAJ**
Batch: 2CSE37

CANDIDATE'S DECLARATION

I, Pratyaksh Bhardwaj do hereby declare that the internship report titled **“Online Course & Learning Management System”** has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature:

Name: Pratyaksh Bhardwaj

Roll No.: 25SCS1003000700

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature:

Name: Pratyaksh Bhardwaj

Roll No.: 25SCS1003000700

INTERNSHIP COMPLETION CERTIFICATE



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Pratyaksh Bhardwaj

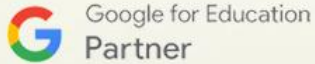
From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Full Stack Developer Intern

Conducted from 20/07/2026 to 20/08/2026



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, likely belonging to Dr. Anurag Shrivastava.

Dr. Anurag Shrivastava
Program Manager

4. PROJECT DESCRIPTION

The major project undertaken during my internship at Codec Technologies Pvt. Ltd. was an Online Course and Learning Management System. The project was developed as a full stack web application with the objective of creating a structured digital environment for online learning. The system brings together course-related information, learning resources, user interaction, and application-level data processing within a common platform.

The project was selected because an online learning platform is a suitable practical problem for understanding full stack development. Such a system requires more than a static website: it requires a user interface, navigation and interaction, application logic, structured data, validation, and communication between the different layers of the application. Developing this project therefore provided exposure to several important stages of software development.

The following sections describe the organization, problem addressed, objectives, scope, technologies and tools, architecture, methodology, expected outcomes, and the work carried out during the internship. The technical descriptions are expanded to provide a complete academic account of the project.

4.1 Introduction

Digital education has changed the way learners access educational material. Instead of depending entirely on physical classrooms and printed resources, learners can use web-based platforms to access course information and learning content from different locations and at convenient times. A Learning Management System provides a structured way of organizing such digital learning activities.

The Online Course and Learning Management System developed during the internship was designed around this idea. The application provides a centralized environment in which courses can be presented in an organized manner and users can interact with

learning-related content through a web interface. The project combines presentation, processing, and data management so that the application behaves as an integrated system.

From the development perspective, the project was useful because it introduced the complete flow of a web request. A user performs an action through the frontend, the request is processed by the application layer, relevant data is accessed or updated, and the result is returned to the interface. Understanding this flow is one of the most important foundations of full stack development.

The project also provided practical exposure to responsive interface design, component organization, form handling, validation, API-based communication, database operations, error handling, testing, and debugging. These activities helped connect programming concepts with the requirements of a usable application.

4.2 Organization Profile

Codec Technologies Pvt. Ltd. was the organization where I completed my internship. The internship certificate supplied for this report confirms that the internship was a one-month AICTE & ICAC approved internship program conducted from 20/07/2026 to 20/08/2026. The certificate identifies my role as Full Stack Developer Intern.

The internship provided a project-oriented environment in which I could apply computer science concepts to web application development. My primary project involved developing an Online Course and Learning Management System. This allowed me to gain practical understanding of how a software project is approached from requirement understanding through implementation and testing.

As an intern, my learning was not limited to writing code. I also had to understand user requirements, divide the project into manageable modules, maintain consistency between frontend and backend behavior, identify errors, test functionality, and document the resulting work. These activities are important in professional software development because a successful application depends on both technical implementation and systematic development practices.

4.3 Problem Statement

The primary problem addressed by the project is the need for a centralized and structured web platform for online courses and learning resources. When educational material is distributed across unrelated pages, documents, messages, or platforms, learners may find

it difficult to understand the course structure, locate required content, and follow a consistent learning path.

The proposed system addresses this challenge by bringing course information and learning content into a single application. The platform is designed to provide an organized interface in which users can browse available learning material and interact with the system through clear navigation and forms.

From the technical perspective, the problem also involves integrating multiple application layers. The frontend must present information clearly, the backend must process requests correctly, and the database must store information in a structured manner. The system therefore represents a typical full stack development problem in which functionality depends on successful communication between several components.

The project also considers common web application concerns such as input validation, error handling, responsive layout, data consistency, and maintainability. These concerns are important because an application can be functionally correct but still provide a poor user experience if it is difficult to navigate or unreliable.

4.4 Project Objectives

The project objectives were defined to address both the functional requirements of the learning platform and the learning goals of the internship.

- To understand the complete software development lifecycle through a practical full stack project.
- To design a clean and user-friendly interface for an online course and learning environment.

- To organize course information and learning resources into a structured digital platform.
- To develop reusable frontend components and consistent page layouts.
- To implement backend application logic for processing user requests and managing system operations.
- To connect the frontend with backend services through HTTP/API-based communication.
- To maintain course and user-related information using structured database operations.
- To apply form validation and basic input handling to improve data quality and application reliability.
- To test application functionality and identify errors through iterative debugging.
- To improve understanding of responsive web design and cross-device usability.
- To practice writing modular, readable, and maintainable code.
- To gain confidence in developing and documenting a complete web-based software project.

4.5 Scope of the Project

The scope of the Online Course and Learning Management System includes the development of the major components required for a web-based learning platform. The system is intended to provide users with a structured way of viewing courses and accessing learning-related information.

At the frontend level, the scope includes a responsive interface, navigation, course presentation, forms, user interaction, and feedback elements. At the application level, the scope includes request handling, validation, business rules, and communication with the data layer. At the data level, the scope includes structured storage and retrieval of relevant information.

A practical implementation can include modules such as user registration and login, course listing, course details, lesson/content organization, learner enrollment, progress tracking, and an administrative area for maintaining courses. These modules form a logical extension of the project theme; the exact features should be matched with the implemented application before final academic submission.

The project can also be extended in the future. Possible extensions include online assessments and quizzes, certificates, instructor dashboards, learner analytics, notifications, search and filtering, payment integration for paid courses, recommendation systems, discussion forums, and cloud deployment. These features are outside the minimum scope described here and are identified as future enhancement possibilities.

4.6 Technologies and Tools Used

The project was approached using a modern full stack web development architecture. The following stack provides a practical and coherent implementation for an Online Course and Learning Management System. Because the internship certificate itself does not list individual frameworks or software tools, the technologies in this section should be treated as the intended/representative project stack and verified against the actual source code before submission.

Technology / Tool	Purpose and Role
HTML5	Provides the structural foundation of web pages and semantic organization of content.
CSS3	Used for layout, typography, spacing, responsive design, and visual presentation.
JavaScript (ES6+)	Provides client-side logic, interaction, event handling, validation, and dynamic behavior.
React.js	Can be used to build reusable, component-based frontend interfaces and manage UI state.
React Router	Supports navigation between pages/views without unnecessary full-page reloads.
Bootstrap / CSS Responsive Utilities	Can assist with responsive layouts, grids, forms, navigation, and consistent UI components.
Node.js	Provides a JavaScript runtime for implementing server-side application logic.
Express.js	Can be used to create REST-style API endpoints, middleware, routing, and request handling.
MongoDB	Can be used as a document-oriented database for users, courses, lessons, enrollments, and progress information.
Mongoose	Provides a structured way to define MongoDB schemas and interact with stored documents.
REST API	Defines the communication layer through which frontend components can request or submit application data.
JSON	Used as a lightweight data-interchange format

	between client and server.
Git	Used for version control, tracking changes, and maintaining development history.
GitHub	Can be used for remote source-code hosting, collaboration, and repository management.
Visual Studio Code	A practical code editor for writing, organizing, and debugging the project.
Postman	Can be used to test backend API endpoints independently from the frontend.
Web Browser Developer Tools	Useful for inspecting elements, debugging JavaScript, checking network requests, and analyzing responsive layouts.

The use of these technologies supports a separation of concerns. HTML/CSS and the frontend framework handle presentation and interaction, Node.js/Express.js handle application processing and APIs, and MongoDB provides persistent data storage. Version control and development tools support the engineering workflow around the application.

4.6.1 Frontend Technologies

The frontend is the part of the system directly visible to the learner. A learning platform requires a clear and responsive interface because users may access course information from desktops, laptops, tablets, or mobile devices. HTML5 can provide semantic page structure, while CSS3 controls layout, spacing, typography, and responsive behavior.

JavaScript provides interactive behavior such as form validation, button actions, dynamic content updates, and communication with backend services. A component-based framework such as React.js is suitable for the project because common interface elements—navigation bars, course cards, forms, buttons, lesson views, and dashboards—can be developed as reusable components.

Reusable components reduce duplication and make the interface easier to maintain. For example, a single CourseCard component can display different courses by receiving course information as properties. Similarly, a common navigation component can be used across multiple screens, improving visual consistency.

4.6.2 Backend Technologies

The backend is responsible for processing requests received from the frontend and applying the application's rules. Node.js provides a JavaScript runtime that allows server-side development, while Express.js can simplify routing and middleware management.

The backend can expose endpoints for operations such as user registration, authentication, course retrieval, course creation, enrollment, progress updates, and other application functions. Each endpoint should validate incoming data and return an appropriate response so that the frontend can present the result to the user.

Middleware can be used for common concerns such as parsing JSON requests, handling authentication, validating input, logging requests, and managing errors. This helps keep the application modular and makes individual routes easier to understand.

4.6.3 Database Technology

A learning platform needs persistent storage because information should remain available after a user closes the browser. MongoDB can be used to store documents representing entities such as users, courses, lessons, enrollments, and progress records.

A suitable data model should avoid unnecessary duplication while still allowing efficient retrieval. For example, course documents may contain course metadata while lessons can be represented as separate related documents or embedded structures depending on the design. Mongoose can be used to define schemas and provide validation and structured database access.

Database operations include creating records, reading records, updating information, and deleting records when required. These operations form the basis of CRUD functionality in the learning management system.

4.6.4 Development and Testing Tools

Visual Studio Code can be used as the primary development environment because it provides source-code editing, project navigation, extensions, integrated terminal access, and debugging support. Git helps track changes and provides a way to return to earlier versions when a modification introduces a problem.

Postman is useful for backend development because API endpoints can be tested separately from the frontend. Browser developer tools are also important for identifying frontend errors, inspecting network requests, checking HTML/CSS behavior, and analyzing console messages.

4.7 System Architecture

The system follows a layered architecture in which responsibilities are separated into presentation, application, and data layers. This approach makes the application easier to understand, test, maintain, and extend.

Layer	Main Responsibility	Typical Components
Presentation Layer	Displays information and accepts user interaction.	React components, HTML5, CSS3, JavaScript, responsive UI
Application Layer	Processes requests, applies business logic, validates data, and manages APIs.	Node.js, Express.js, controllers, middleware, REST endpoints
Data Layer	Stores and retrieves persistent application information.	MongoDB, Mongoose schemas/models
Development Layer	Supports source management and testing.	Git, GitHub, VS Code, Postman, browser developer tools

Figure 4.1: Conceptual architecture of the Online Course and Learning Management System.

USER	PRESENTATION LAYER	APPLICATION LAYER	DATA LAYER
Learner / Admin	Frontend UI & Components	REST APIs & Business Logic	Database & Models
Input / Interaction	Validation & Rendering	Processing / Authentication	Create / Read / Update / Delete

A typical request begins when the user performs an action in the interface. The frontend validates basic information and sends a request to the backend. The backend receives the request, verifies it, applies business rules, and communicates with the database if data must be read or changed. The backend then sends a response, commonly in JSON format, and the frontend updates the interface based on the result.

This separation is beneficial because changes in one layer can often be made without rewriting the entire application. For example, a visual change in the course card can be implemented in the frontend without changing the database structure, while a database optimization can be performed without redesigning the user interface.

4.8 Methodology

The development methodology followed an iterative approach. Instead of attempting to build the entire platform at once, the project can be divided into smaller modules and developed progressively. Each stage produces a clearer understanding of the requirements and allows issues to be identified early.

4.8.1 Requirement Analysis

The first step was to understand what the learning platform should accomplish. The core requirements were identified around course presentation, learning content, user interaction, and data management. Functional requirements describe what the system should do, while non-functional requirements describe qualities such as usability, responsiveness, reliability, maintainability, and performance.

4.8.2 System Planning

After requirements were understood, the application was divided into logical modules. Planning the modules before implementation helps reduce complexity and establishes a clear relationship between screens, APIs, and database entities.

4.8.3 UI and Component Design

The user interface was planned around simple navigation and clear presentation of course information. Reusable components are useful for repeated elements such as navigation bars, course cards, buttons, forms, headers, footers, and content sections. Responsive design was considered so that the interface can adapt to different screen sizes.

4.8.4 Frontend Implementation

The frontend implementation focuses on converting the planned design into functional pages and components. JavaScript handles interaction, while a component-based approach allows individual sections to be developed and tested independently. Form fields can include validation to reduce incorrect input.

4.8.5 Backend and API Development

The backend provides the services required by the frontend. API routes can be organized according to resources such as users and courses. Controllers or route handlers process requests, perform validation, interact with the database, and return responses.

4.8.6 Database Integration

Database integration connects application logic with persistent information. Schemas define the expected structure of stored records, while CRUD operations support normal application workflows. Data validation at the application and database levels helps reduce inconsistent records.

4.8.7 Testing and Debugging

Testing was an important part of the development process. Individual functions can first be checked separately, followed by integration testing in which frontend, backend, and database operations are used together. Errors discovered during testing are analyzed, corrected, and tested again.

4.8.8 Final Refinement and Documentation

After core functionality is available, the application can be refined by improving layout consistency, error messages, navigation, responsiveness, and code organization. Documentation records the purpose of the project, its architecture, technology choices, and learning outcomes.

4.8.9 Testing Approach

Test Area	Example Test	Expected Result
Navigation	Open major pages and use navigation controls.	Correct page/view is displayed.
Forms	Submit empty or invalid fields.	Validation identifies missing/incorrect information.
Authentication	Enter valid and invalid login information.	Valid credentials are accepted and invalid input is rejected.
Course Listing	Request available course data.	Relevant course information is returned and displayed.
Course Details	Open a selected course.	Correct course details and learning

		content are displayed.
API	Send requests through an API testing tool.	Server returns appropriate status and response data.
Database	Create, retrieve, update, and delete test records.	Database operations complete correctly.
Responsive UI	Open the application at different viewport sizes.	Layout remains usable and readable.
Error Handling	Trigger invalid requests or unavailable data.	User receives an understandable error response.

4.9 Expected Outcomes

The expected software outcome is a structured web-based learning platform through which users can interact with online course information and learning resources. The system should demonstrate successful integration of the frontend, backend, and data layers.

From the user's perspective, the platform should provide clear navigation, understandable content organization, and predictable interactions. From the developer's perspective, the project should demonstrate modularity, separation of concerns, structured data handling, and an organized development process.

The internship also has important learning outcomes. By working on the project, I developed a better understanding of how programming concepts are applied in a software project. I gained practical exposure to debugging and testing and learned that software development is an iterative activity in which implementation is continuously improved based on testing and feedback.

Outcome	Description
Functional Learning Platform	A centralized environment for presenting and managing online learning content.
Full Stack Understanding	Practical understanding of frontend, backend, API, and database interaction.
Improved Problem Solving	Experience in breaking larger problems into smaller development tasks.
Testing Skills	Ability to identify, reproduce, and correct functional and interface issues.
Code Organization	Understanding of reusable components, modular code, and separation of responsibilities.
Professional Skills	Improved technical documentation, project communication, and development discipline.

5. BIBLIOGRAPHY / REFERENCES

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